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Java SE 27 & JDK 27

Class Modifier

java.lang.Object
    java.lang.reflect.Modifier

```
public final class Modifier
extends Object
```

Provides `static` methods and constants to decode classfile access and property flags with corresponding Java language modifiers.

Modifier interpretation is context-sensitive: for example, the `isSynchronized` check is only meaningful for method access flags, representing the `synchronized` modifier on methods. A `true` return on a field access flags does not indicate that field has the `synchronized` modifier.

API Note:

The mappings from classfile access flags to Java language modifiers have diverged during the evolution of the Java SE Platform. Many access flags and Java language modifiers are not represented in this class; the mappings represented in this class are not sufficient to reconstruct Java langugage modifiers from access flags, and vice versa.

Since:

1.1

See Also:

`AccessFlag`,
`Modifier`,
Java programming language and JVM modeling in core reflection,
`Class.getModifiers()`,
`Member.getModifiers()`,
`Parameter.getModifiers()`

Field Summary

Fields

Modifier and Type	Field	Description
`static final int`	**`ABSTRACT`**	The `int` value representing the `abstract` modifier.
`static final int`	**`FINAL`**	The `int` value representing the `final` modifier.
`static final int`	**`INTERFACE`**	

	The `int` value representing the `interface` modifier.
`static final int` **`NATIVE`**	The `int` value representing the `native` modifier.
`static final int` **`PRIVATE`**	The `int` value representing the `private` modifier.
`static final int` **`PROTECTED`**	The `int` value representing the `protected` modifier.
`static final int` **`PUBLIC`**	The `int` value representing the `public` modifier.
`static final int` **`STATIC`**	The `int` value representing the `static` modifier.
`static final int` **`STRICT`**	The `int` value representing the `strictfp` modifier.
`static final int` **`SYNCHRONIZED`**	The `int` value representing the `synchronized` modifier.
`static final int` **`TRANSIENT`**	The `int` value representing the `transient` modifier.
`static final int` **`VOLATILE`**	The `int` value representing the `volatile` modifier.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
		Deprecated.

static int	**classModifiers**()	This method was originally created to support the now-deprecated Modifier::toString(int) method.
static int	**constructorModifiers**()	**Deprecated.** This method was originally created to support the now-deprecated Modifier::toString(int) method.
static int	**fieldModifiers**()	**Deprecated.** This method was originally created to support the now-deprecated Modifier::toString(int) method.
static int	**interfaceModifiers**()	**Deprecated.** This method was originally created to support the now-deprecated Modifier::toString(int) method.
static boolean	**isAbstract**(int mod)	Return `true` if the integer argument includes the `abstract` modifier, `false` otherwise.
static boolean	**isFinal**(int mod)	Return `true` if the integer argument includes the `final` modifier, `false` otherwise.
static boolean	**isInterface**(int mod)	Return `true` if the integer argument

		includes the `interface` modifier, `false` otherwise.
`static boolean`	**`isNative`**`(int mod)`	Return `true` if the integer argument includes the `native` modifier, `false` otherwise.
`static boolean`	**`isPrivate`**`(int mod)`	Return `true` if the integer argument includes the `private` modifier, `false` otherwise.
`static boolean`	**`isProtected`**`(int mod)`	Return `true` if the integer argument includes the `protected` modifier, `false` otherwise.
`static boolean`	**`isPublic`**`(int mod)`	Return `true` if the integer argument includes the `public` modifier, `false` otherwise.
`static boolean`	**`isStatic`**`(int mod)`	Return `true` if the integer argument includes the `static` modifier, `false` otherwise.
`static boolean`	**`isStrict`**`(int mod)`	Return `true` if the integer argument includes the `strictfp` modifier, `false` otherwise.
`static boolean`	**`isSynchronized`**`(int mod)`	Return `true` if the integer argument includes the

		`synchronized` modifier, `false` otherwise.
`static boolean`	**`isTransient`**`(int mod)`	Return `true` if the integer argument includes the `transient` modifier, `false` otherwise.
`static boolean`	**`isVolatile`**`(int mod)`	Return `true` if the integer argument includes the `volatile` modifier, `false` otherwise.
`static int`	**`methodModifiers`**`()`	**Deprecated.** This method was originally created to support the now-deprecated `Modifier::toString(int)` method.
`static int`	**`parameterModifiers`**`()`	**Deprecated.** This method was originally created to support the now-deprecated `Modifier::toString(int)` method.
`static String`	**`toString`**`(int mod)`	**Deprecated.** Modifier interpretation is context-sensitive; this API may report an incomplete or incorrect list of Java language modifiers.

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Field Details

PUBLIC

`public static final int PUBLIC`

The `int` value representing the `public` modifier.

See Also:

`AccessFlag.PUBLIC`, Constant Field Values

PRIVATE

`public static final int PRIVATE`

The `int` value representing the `private` modifier.

See Also:

`AccessFlag.PRIVATE`, Constant Field Values

PROTECTED

`public static final int PROTECTED`

The `int` value representing the `protected` modifier.

See Also:

`AccessFlag.PROTECTED`, Constant Field Values

STATIC

`public static final int STATIC`

The `int` value representing the `static` modifier.

See Also:

`AccessFlag.STATIC`, Constant Field Values

FINAL

`public static final int FINAL`

The `int` value representing the `final` modifier.

See Also:

AccessFlag.FINAL, Constant Field Values

SYNCHRONIZED

```
public static final int SYNCHRONIZED
```

The int value representing the synchronized modifier.

See Also:

AccessFlag.SYNCHRONIZED, Constant Field Values

VOLATILE

```
public static final int VOLATILE
```

The int value representing the volatile modifier.

See Also:

AccessFlag.VOLATILE, Constant Field Values

TRANSIENT

```
public static final int TRANSIENT
```

The int value representing the transient modifier.

See Also:

AccessFlag.TRANSIENT, Constant Field Values

NATIVE

```
public static final int NATIVE
```

The int value representing the native modifier.

See Also:

AccessFlag.NATIVE, Constant Field Values

INTERFACE

```
public static final int INTERFACE
```

The int value representing the interface modifier.

See Also:

AccessFlag.INTERFACE, Constant Field Values

ABSTRACT

```
public static final int ABSTRACT
```

The `int` value representing the `abstract` modifier.

See Also:

`AccessFlag.ABSTRACT`, Constant Field Values

STRICT

```
public static final int STRICT
```

The `int` value representing the `strictfp` modifier.

See Also:

`AccessFlag.STRICT`, Constant Field Values

Method Details

isPublic

```
public static boolean isPublic(int mod)
```

Return `true` if the integer argument includes the `public` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `public` modifier; `false` otherwise.

isPrivate

```
public static boolean isPrivate(int mod)
```

Return `true` if the integer argument includes the `private` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `private` modifier; `false` otherwise.

isProtected

```
public static boolean isProtected(int mod)
```

Return `true` if the integer argument includes the `protected` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `protected` modifier; `false` otherwise.

isStatic

```
public static boolean isStatic(int mod)
```

Return `true` if the integer argument includes the `static` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `static` modifier; `false` otherwise.

isFinal

```
public static boolean isFinal(int mod)
```

Return `true` if the integer argument includes the `final` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `final` modifier; `false` otherwise.

isSynchronized

```
public static boolean isSynchronized(int mod)
```

Return `true` if the integer argument includes the `synchronized` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `synchronized` modifier; `false` otherwise.

isVolatile

```
public static boolean isVolatile(int mod)
```

Return `true` if the integer argument includes the `volatile` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `volatile` modifier; `false` otherwise.

isTransient

```
public static boolean isTransient(int mod)
```

Return `true` if the integer argument includes the `transient` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `transient` modifier; `false` otherwise.

isNative

```
public static boolean isNative(int mod)
```

Return `true` if the integer argument includes the `native` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `native` modifier; `false` otherwise.

isInterface

```
public static boolean isInterface(int mod)
```

Return `true` if the integer argument includes the `interface` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `interface` modifier; `false` otherwise.

isAbstract

```
public static boolean isAbstract(int mod)
```

Return `true` if the integer argument includes the `abstract` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `abstract` modifier; `false` otherwise.

isStrict

```
public static boolean isStrict(int mod)
```

Return `true` if the integer argument includes the `strictfp` modifier, `false` otherwise.

Parameters:

`mod` - a set of modifiers

Returns:

`true` if `mod` includes the `strictfp` modifier; `false` otherwise.

toString

```
@Deprecated(since="27")
public static String toString(int mod)
```

Deprecated.
Modifier interpretation is context-sensitive; this API may report an incomplete or incorrect list of Java language modifiers. The mappings from `class` file access flags to Java language modifiers have diverged during the evolution of the Java SE Platform.

Use `AccessFlag` to examine access flags; `toGenericString` methods on reflective objects print Java language modifiers.

Return a string describing the access modifier flags in the specified modifier. For example:

```
public final synchronized strictfp
```

The modifier names are returned in an order consistent with the suggested modifier orderings given in sections 8.1.1, 8.3.1, 8.4.3, 8.8.3, and 9.1.1 of *The Java Language Specification*. The full modifier ordering used by this method is:

```
public protected private abstract static final transient
volatile synchronized native strictfp interface
```

The `interface` modifier discussed in this class is not a true modifier in the Java language and it appears after all other modifiers listed by this method. This method may return a string of modifiers that are not valid modifiers of a Java entity; in other words, no checking is done on the possible validity of the combination of modifiers represented by the input. This method also omits all access flags without a corresponding source modifier.

Parameters:

`mod` - a set of modifiers

Returns:

a string representation of the set of modifiers represented by `mod`

classModifiers

```
@Deprecated(since="27")
public static int classModifiers()
```

Deprecated.
This method was originally created to support the now-deprecated `Modifier::toString(int)` method. Use `AccessFlag.Location` to inspect structure-specific access modifier properties.

Return an `int` value OR-ing together the source language modifiers that can be applied to a class.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to a class.

See *Java Language Specification*:

8.1.1 Class Modifiers

Since:

1.7

See Also:

`AccessFlag.Location.CLASS`,
`AccessFlag.Location.INNER_CLASS`

interfaceModifiers

```
@Deprecated(since="27")
public static int interfaceModifiers()
```

Deprecated.
This method was originally created to support the now-deprecated `Modifier::toString(int)` method. Use `AccessFlag.Location` to inspect structure-specific access modifier properties.

Return an `int` value OR-ing together the source language modifiers that can be applied to an interface.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to an interface.

See *Java Language Specification*:

9.1.1 Interface Modifiers

Since:

1.7

See Also:

`AccessFlag.Location.CLASS`,
`AccessFlag.Location.INNER_CLASS`

constructorModifiers

```
@Deprecated(since="27")
public static int constructorModifiers()
```

Deprecated.
This method was originally created to support the now-deprecated `Modifier::toString(int)` method. Use `AccessFlag.Location` to inspect structure-specific access modifier properties.

Return an `int` value OR-ing together the source language modifiers that can be applied to a constructor.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to a constructor.

See *Java Language Specification*:

8.8.3 Constructor Modifiers

Since:

1.7

See Also:

`AccessFlag.Location.METHOD`

methodModifiers

```
@Deprecated(since="27")
public static int methodModifiers()
```

Deprecated.
This method was originally created to support the now-deprecated `Modifier::toString(int)` method. Use `AccessFlag.Location` to inspect structure-specific access modifier properties.

Return an `int` value OR-ing together the source language modifiers that can be applied to a method.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to a method.

See *Java Language Specification*:

8.4.3 Method Modifiers

Since:

1.7

See Also:

`AccessFlag.Location.METHOD`

fieldModifiers

```
@Deprecated(since="27")
public static int fieldModifiers()
```

Deprecated.
This method was originally created to support the now-deprecated `Modifier::toString(int)` method. Use `AccessFlag.Location` to inspect structure-specific access modifier properties.

Return an `int` value OR-ing together the source language modifiers that can be applied to a field.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to a field.

See *Java Language Specification*:

8.3.1 Field Modifiers

Since:

1.7

See Also:

`AccessFlag.Location.FIELD`

parameterModifiers

```
@Deprecated(since="27")
public static int parameterModifiers()
```

Deprecated.
This method was originally created to support the now-deprecated `Modifier::toString(int)` method. Use `AccessFlag.Location` to inspect structure-specific access modifier properties.

Return an `int` value OR-ing together the source language modifiers that can be applied to a parameter.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to a parameter.

See *Java Language Specification*:

8.4.1 Formal Parameters

Since:

1.8

See Also:

`AccessFlag.Location.METHOD_PARAMETER`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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